

사이클별 지표 패널

M02Ch109

각 패널 = 한 지표의 cycle 궤적 (routine 0.5C).

early/late 점선 = 앞·뒤 5포인트 median.

트렌드 = 선형 slope/100cyc + aging 기대방향 대조.

주의: LAM_PE / *_proxy / pattern_score는 가설·proxy이며 절대 LAM%/LLI%가 아닙니다. R_ohmic은 $\sqrt{t}$ 외삽 proxy (0.1s 단일값 ≠ 순수 옴).

트렌드 요약 — M02Ch109

- 유효 지표: 32개 · aging 방향 일치 15 · 반대 1

하락 트렌드 (상위)

- LAM 곡선 proxy: early=-2.01 → late=-12.04 (Δ =-10.03, -4.231%/100cyc) → decreasing · context
- 방전 플래토 폭: early=17.18 → late=14.39 (Δ =-2.783, -1.057Q-units/100cyc) → decreasing · matches_aging
- R Ω 성장률 /100cyc: early=1.097 → late=0.7898 (Δ =-0.3076, -0.3082m Ω /100cyc/100cyc) → decreasing · context
- 방전 플래토 ΔV : early=-0.0316 → late=-0.1041 (Δ =-0.07246, -0.03168V/100cyc) → decreasing · context
- 저SOC 히스테리시스: early=0.08696 → late=0.0429 (Δ =-0.04406, -0.01859V/100cyc) → decreasing · opposite_aging

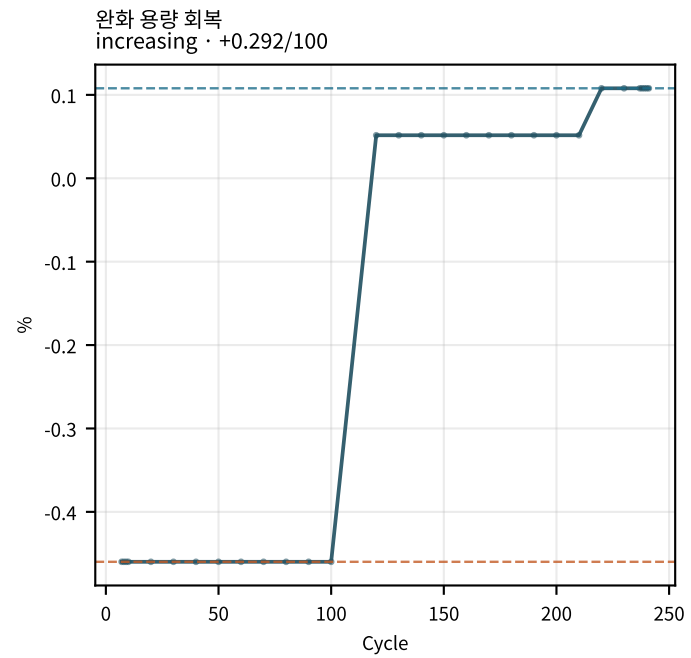
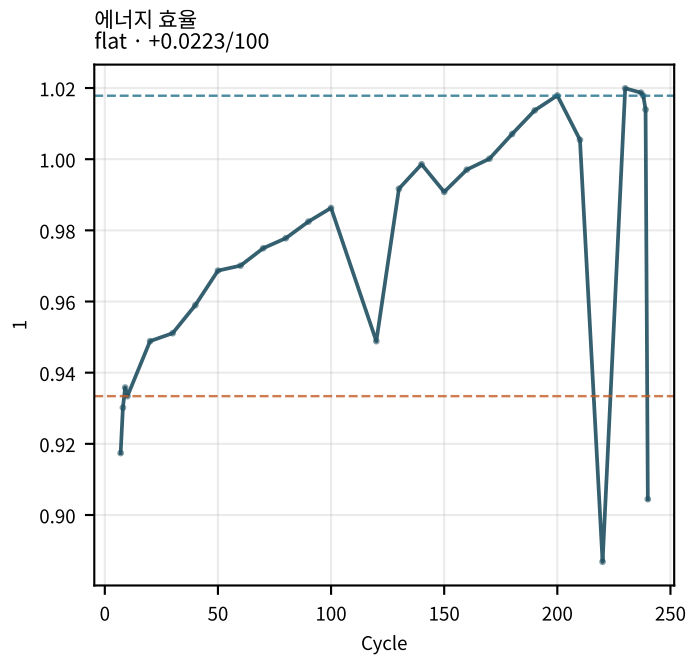
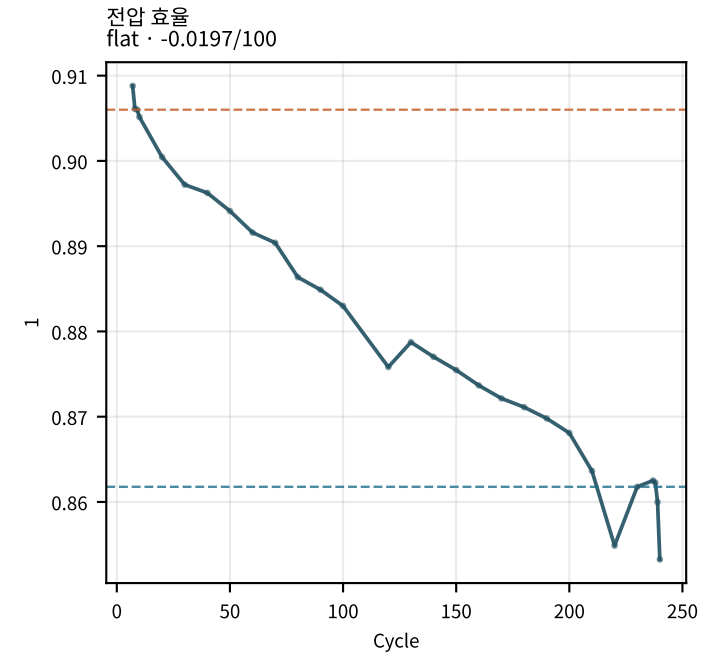
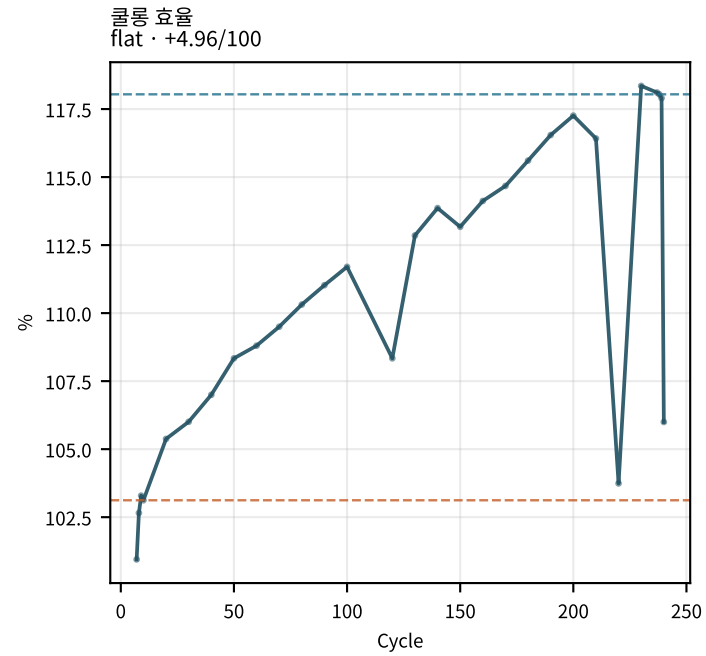
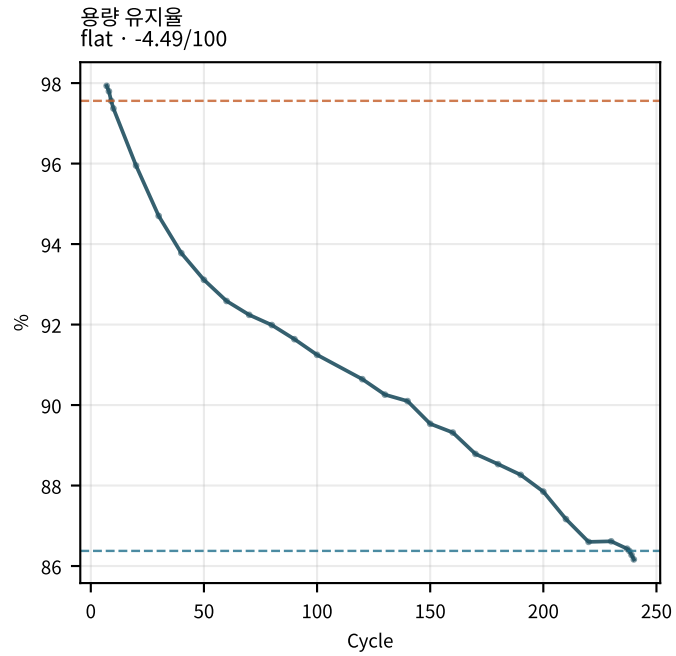
상승 트렌드 (상위)

- 옴 저항 (SOC50): early=1.107 → late=2.765 (Δ =+1.658, +0.7947m Ω /100cyc) → increasing · matches_aging
- 기계/화학 비: early=1.51 → late=3.068 (Δ =+1.559, +0.78761/100cyc) → increasing · matches_aging
- LLI 곡선 proxy: early=-0.08575 → late=1.893 (Δ =+1.979, +0.601%/100cyc) → increasing · context
- 완화 용량 회복: early=-0.4599 → late=0.1079 (Δ =+0.5678, +0.2924%/100cyc) → increasing · matches_aging
- 계단 R 패턴: early=0.4033 → late=0.8515 (Δ =+0.4483, +0.17430-1/100cyc) → increasing · matches_aging

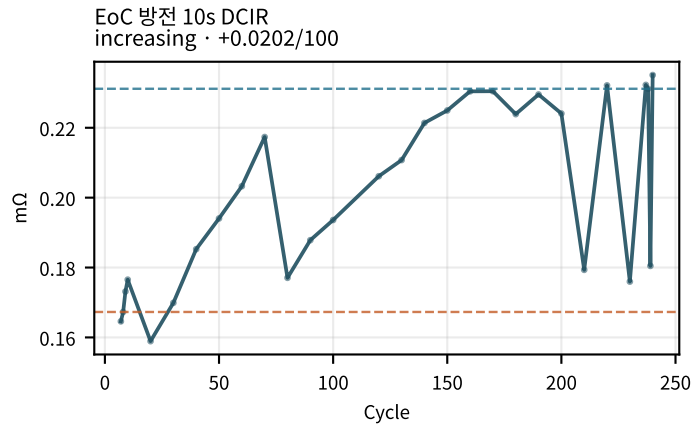
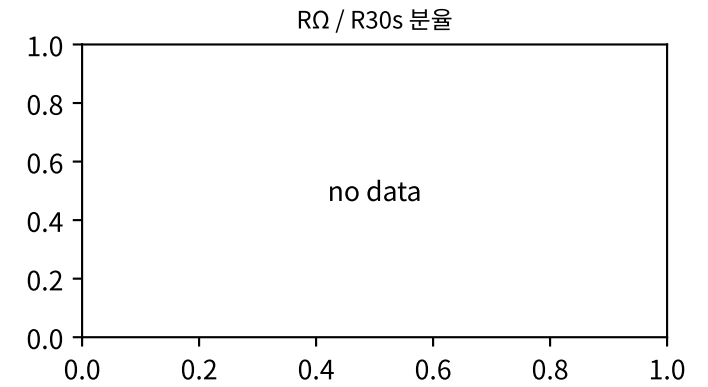
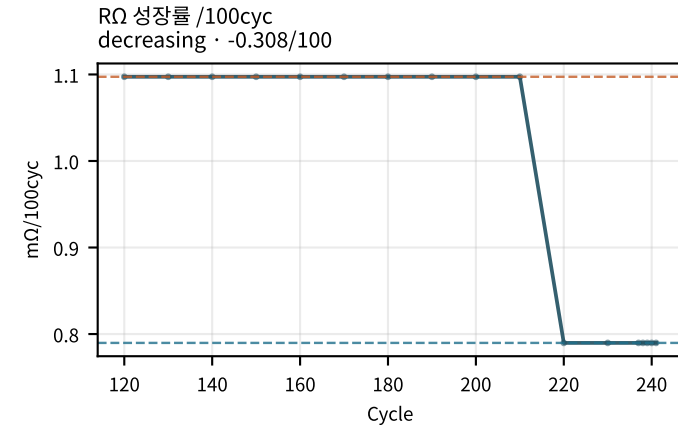
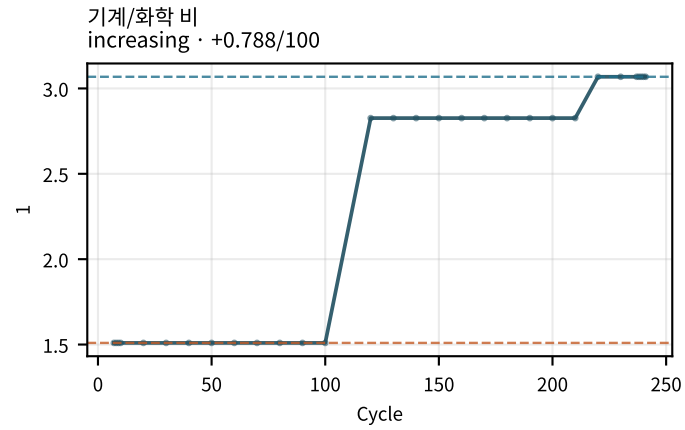
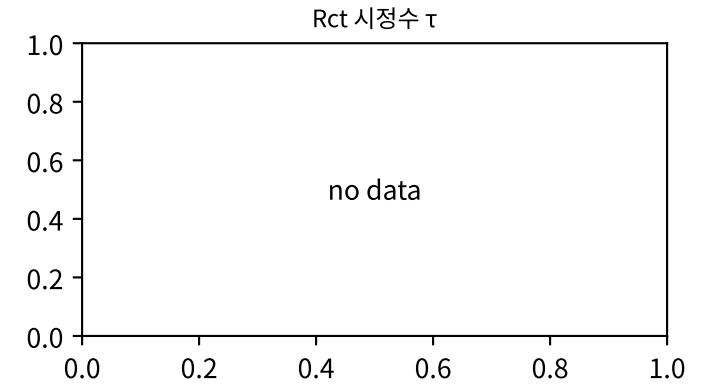
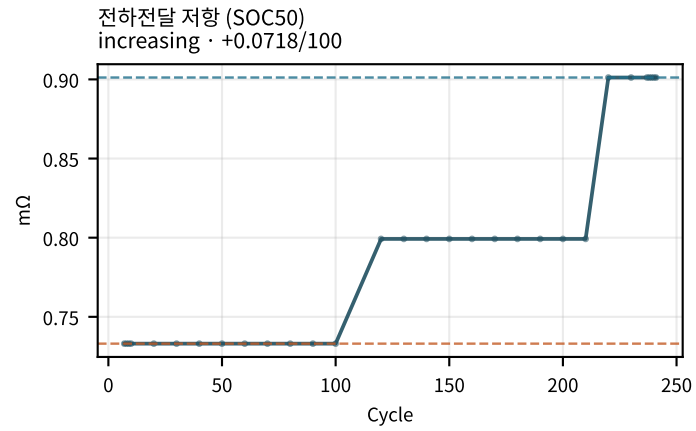
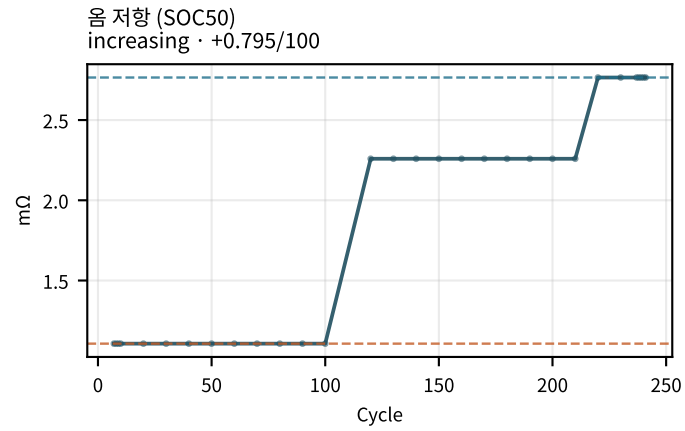
기대 aging과 반대

- 저SOC 히스테리시스: early=0.08696 → late=0.0429 (Δ =-0.04406, -0.01859V/100cyc) → decreasing · opposite_aging

M02Ch109 · 용량 · 효율

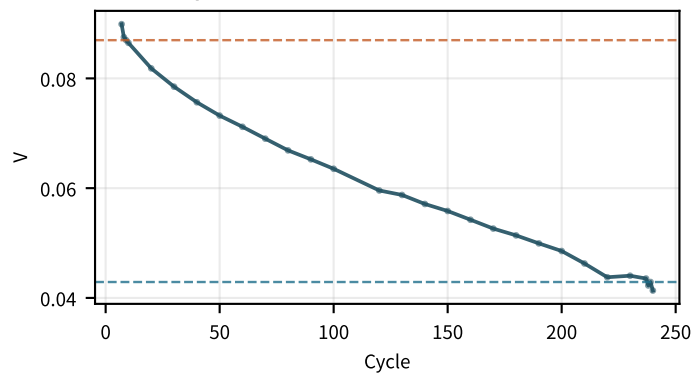


M02Ch109 · 저항 · 분해

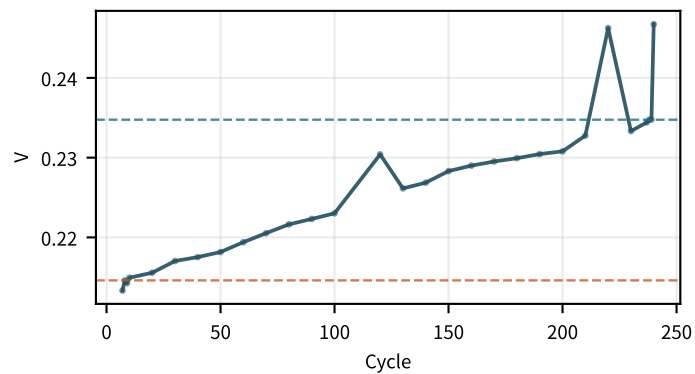


M02Ch109 · 곡선 형상 · dQ/dV

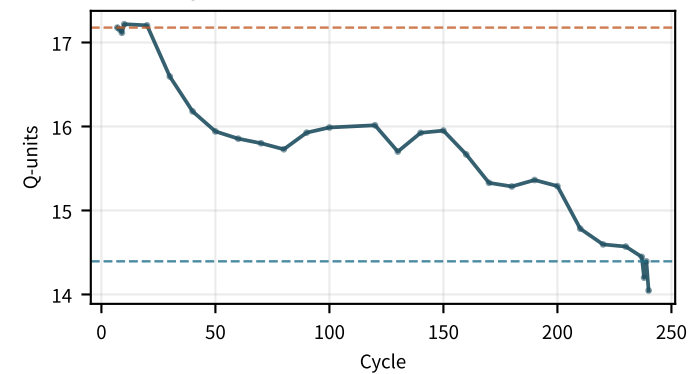
저SOC 히스테리시스
decreasing · $-0.0186/100$



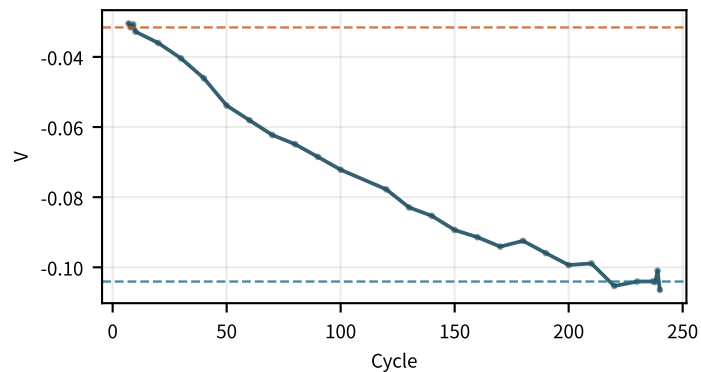
고SOC 히스테리시스
flat · $+0.0101/100$



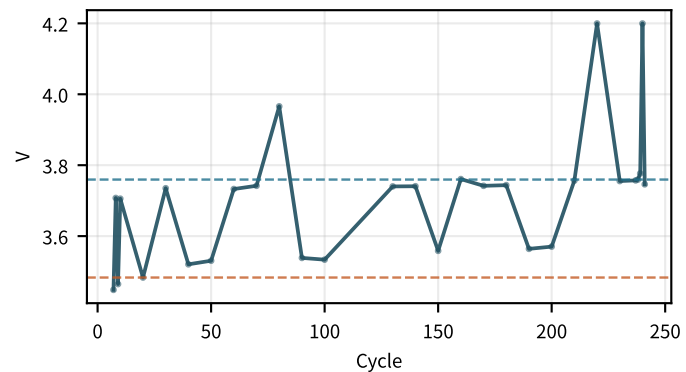
방전 플래토 폭
decreasing · $-1.06/100$



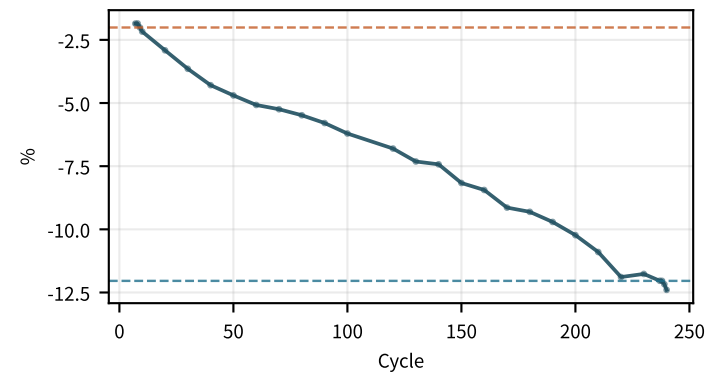
방전 플래토 ΔV
decreasing · $-0.0317/100$



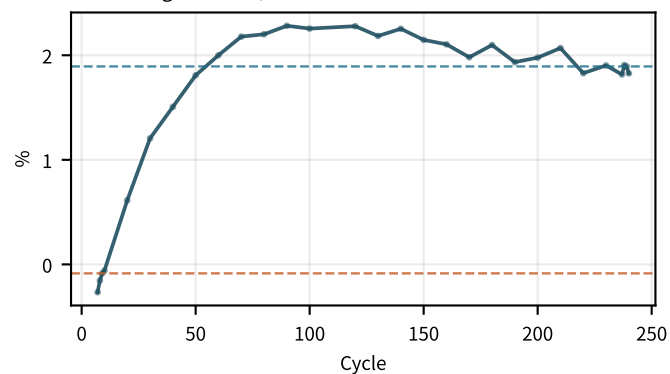
충전 dQ/dV 피크1 V
flat · $+0.108/100$



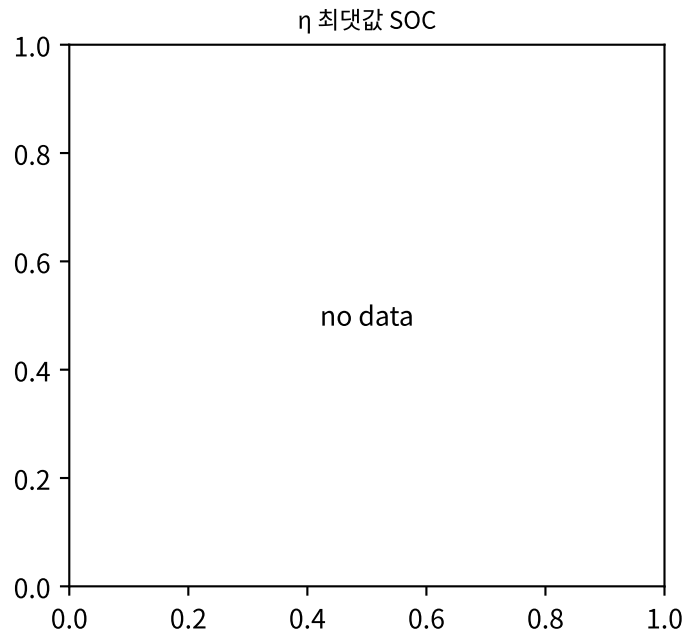
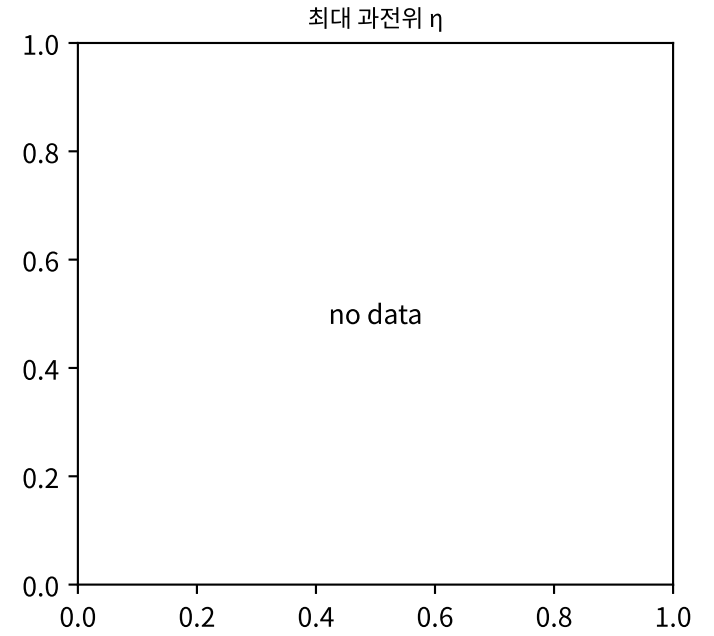
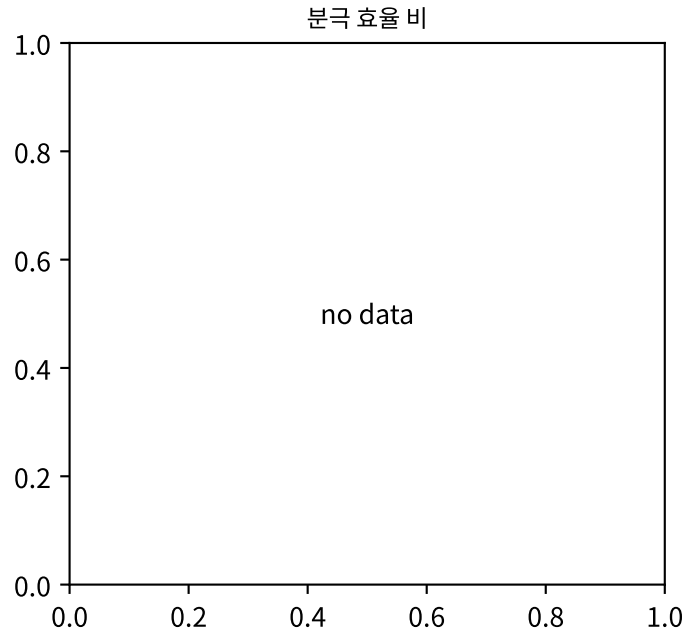
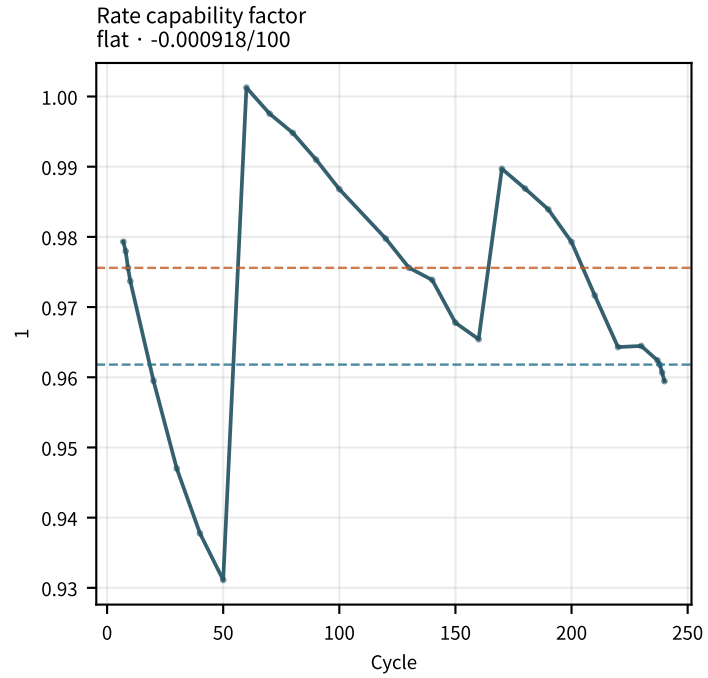
LAM 곡선 proxy
decreasing · $-4.23/100$



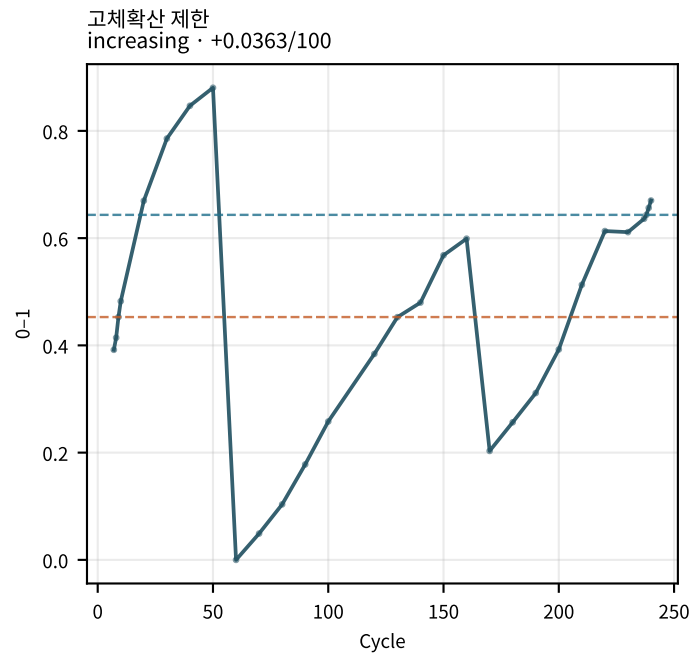
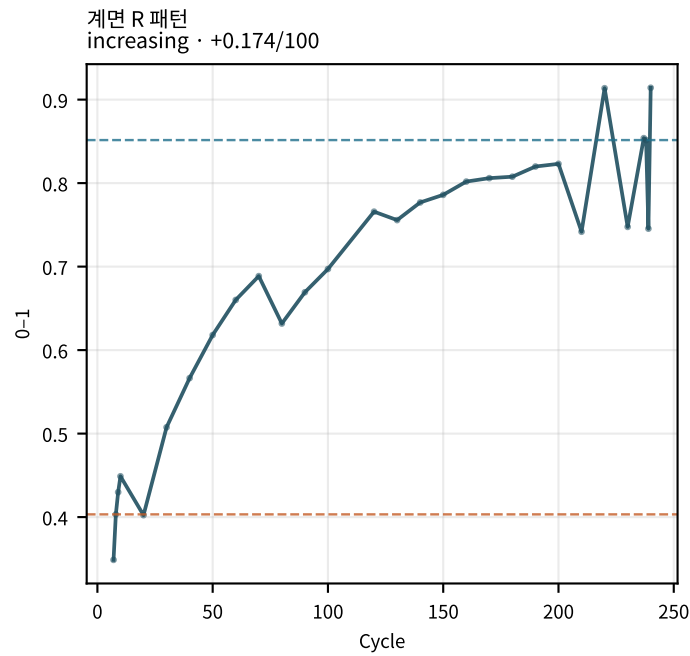
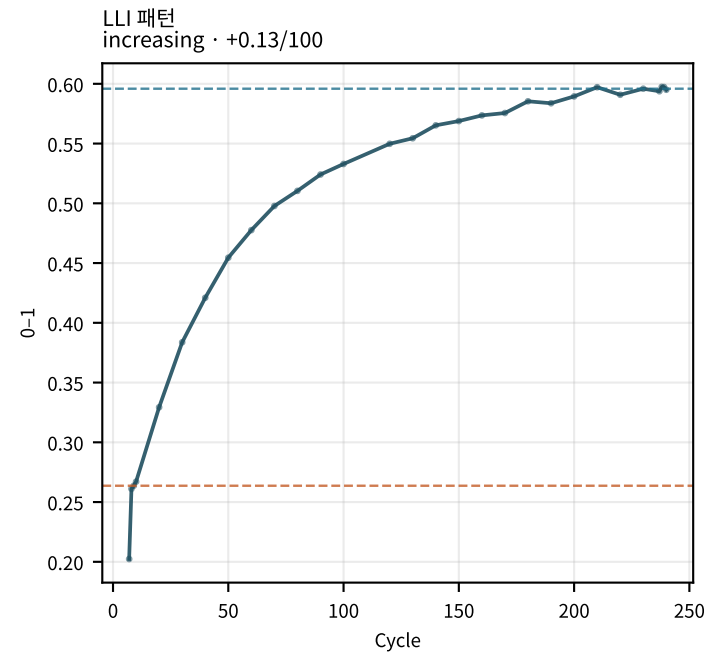
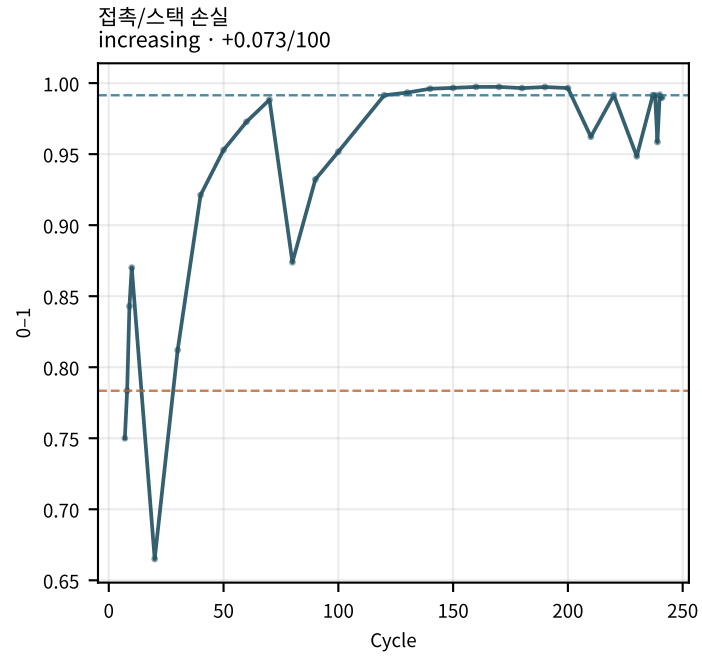
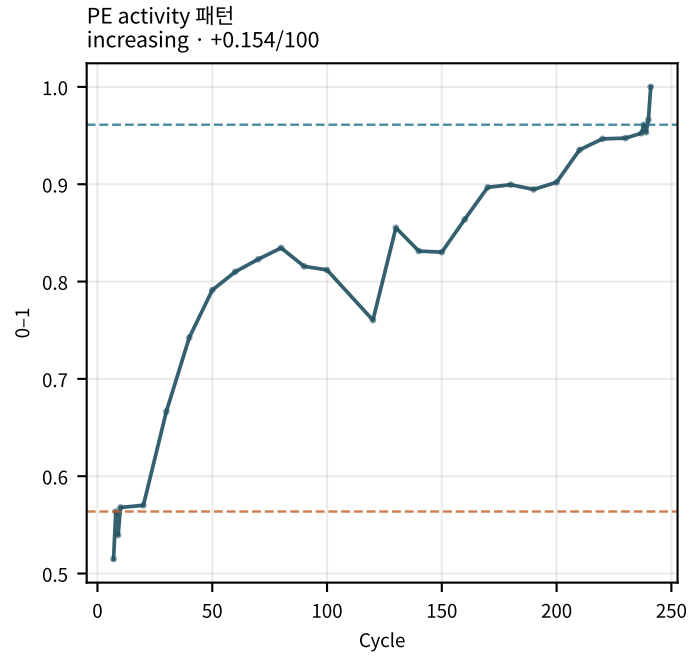
LLI 곡선 proxy
increasing · $+0.601/100$



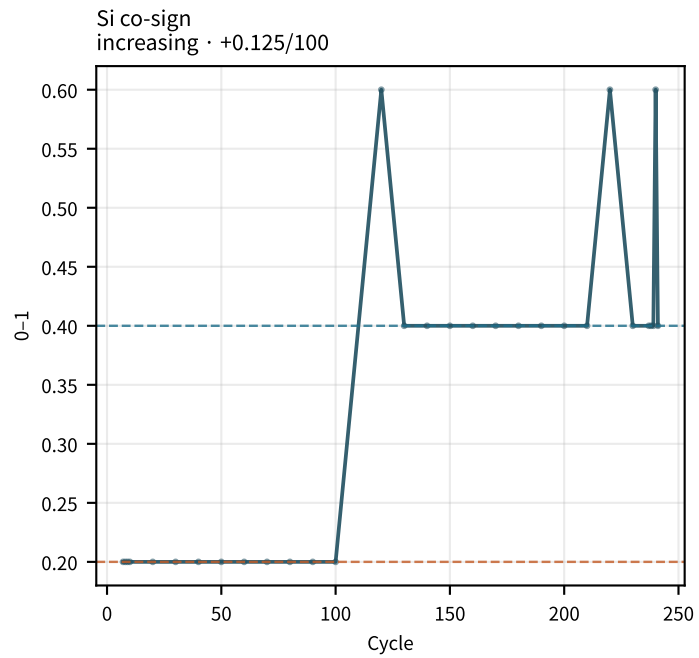
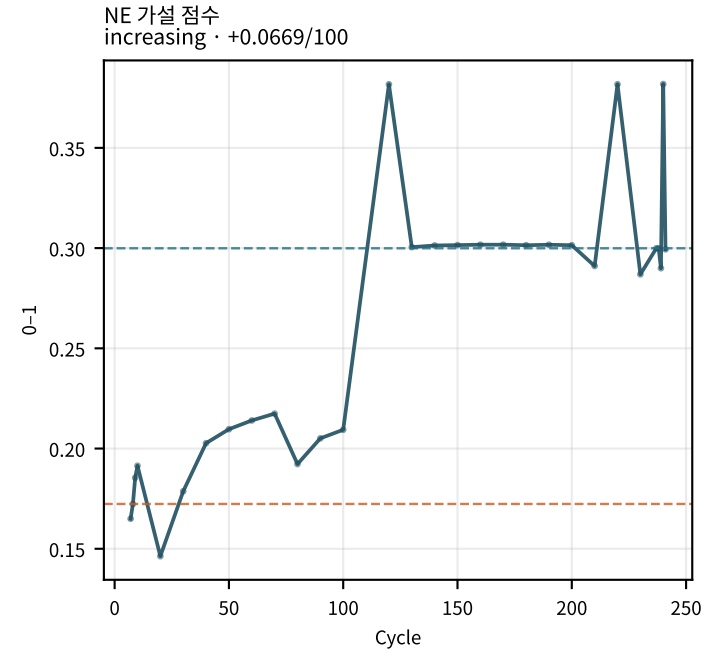
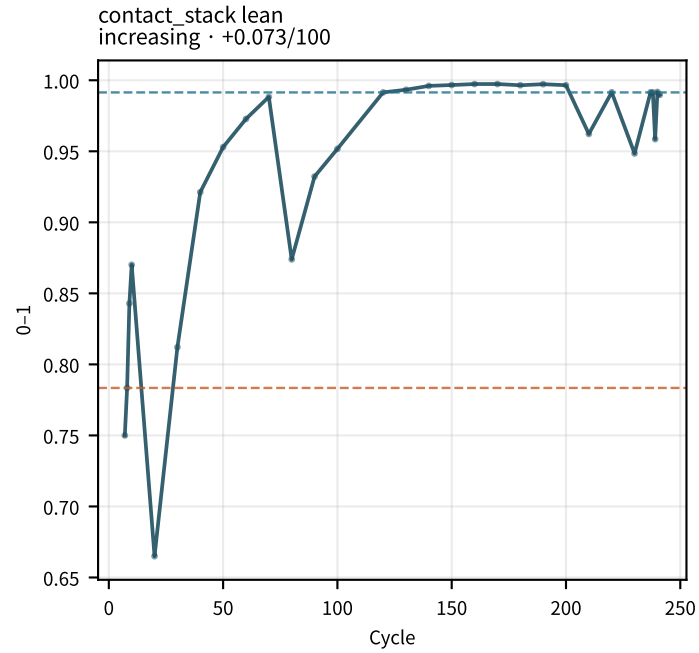
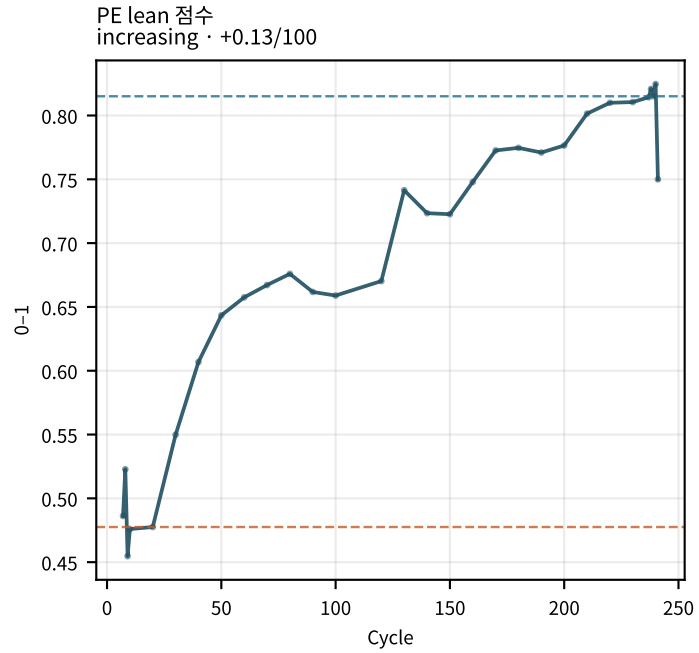
M02Ch109 · 수송 · rate



M02Ch109 · 열화 패턴 점수



M02Ch109 · 전극 lean 가설



지표 설명 · 트렌드 (M02Ch109)

용량 유지율 (SoHQ)

기준 대비 방전용량 유지율. fade/knee의 1차 관측량. | 계산: $Q_dchg(cycle) / Q_baseline \times 100$ (routine 0.5C).

트렌드: 용량 유지율: early=97.56 → late=86.38 ($\Delta=-11.18$, -4.486%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

쿨롱 효율 (CE)

충전 대비 방전용량 비. LLI·부반응 힌트. | 계산: $Q_dchg / Q_chg \times 100$ (이상치 >102% 또는 <85%는 진단에서 제외).

트렌드: 쿨롱 효율: early=103.1 → late=118 ($\Delta=+14.92$, +4.96%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

전압 효율 (VE)

에너지 효율의 전압 측. 저항·분극 증가 시 하락. | 계산: E_dchg / E_chg (Wh).

트렌드: 전압 효율: early=0.906 → late=0.8618 ($\Delta=-0.04423$, -0.019681/100cyc) → flat · stable | aging 기대=decrease | vs=stable

에너지 효율 (EE)

충·방전 에너지 비. | 계산: E_dchg / E_chg .

트렌드: 에너지 효율: early=0.9334 → late=1.018 ($\Delta=+0.08444$, +0.022261/100cyc) → flat · stable | aging 기대=decrease | vs=stable

완화 용량 회복 (Q_relax_pct)

DCIR/RPT 전후 휴지로 회복되는 용량 분율. Si chemo-mech co-sign. | 계산: DCIR 블록 전후 Q 차이 / Q × 100, routine에 forward-fill.

트렌드: 완화 용량 회복: early=-0.4599 → late=0.1079 ($\Delta=+0.5678$, +0.2924%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

옴 저항 (SOC50) (R_ohmic_soc50)

DCIR 펄스 early $\sqrt{t}$ 외삽 $t \rightarrow 0$ 절편. 순수 옴이 아니라 초기 비옴 성분 proxy. | 계산: $R(t)=|\Delta V|/|I|$; $t \in [0, 0.3]$ s 에서 R vs $\sqrt{t}$ 회귀 절편. SOC50 블록 → forward-fill.

트렌드: 옴 저항 (SOC50): early=1.107 → late=2.765 ($\Delta=+1.658$, +0.7947m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch109)

전하전달 저항 (SOC50) (R_ct_soc50)

중간 시간대 잔차의 지수 포화항. 계면/화학 저항 proxy (Cdl 직접 추정 아님). | 계산: $\text{resid} = R - R\Omega - A\sqrt{t}$; $t \in [0.3, 10]s$ 에서 $R_{ct}(1 - e^{-(t/\tau)})$ fit.

트렌드: 전하전달 저항 (SOC50): early=0.7331 → late=0.9012 ($\Delta=+0.1681$, $+0.07175m\Omega/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

Rct 시정수 τ (tau_ct_soc50)

R(t) 잔차 fit의 유효 시정수. $\tau \approx R_{ct} \cdot C_{dl}$ 가설이지만 Cdl은 분리하지 않음. | 계산: curve_fit 초기값 2s, bounds [0.05, 20]; 데이터로 추정.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

기계/화학 비 (mech_vs_chem_ratio)

옴(스택·접촉) vs 전하전달(계면) 상대 비중. | 계산: $R_{ohmic_soc50} / R_{ct_soc50}$.

트렌드: 기계/화학 비: early=1.51 → late=3.068 ($\Delta=+1.559$, $+0.78761/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

R Ω 성장률 /100cyc (R_ohmic_growth_100)

기준 DCIR 대비 100사이클당 옴 저항 증가율. 레벨(R Ω)과 별개 — 감속해도 R Ω 는 계속 상승 가능. | 계산: $(R\Omega - R\Omega_0) / ((cyc - cyc_0)/100)$.

트렌드: R Ω 성장률 /100cyc: early=1.097 → late=0.7898 ($\Delta=-0.3076$, $-0.3082m\Omega/100cyc/100cyc$) → decreasing · context | aging 기대=either | vs=context

R Ω / R30s 분율 (R_ohmic_frac_soc50)

30s 총저항 중 옴 성분 비중. | 계산: $R_{ohmic} / R_{30s_total}$ @ SOC50.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

EoC 방전 10s DCIR (EoC_dchgR_10s)

충전 종료 후 방전 시작 10초 시점 $\Delta V/I$. Rct·확산 포함 총 DCIR. | 계산: $|V_0 - V(10s)|/|I| \times 1000$. 순수 R Ω 아님.

트렌드: EoC 방전 10s DCIR: early=0.1673 → late=0.2312 ($\Delta=+0.06388$, $+0.02024m\Omega/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch109)

저SOC 히스테리시스 (hyst_area_low)

저SOC 충·방전 전압 면적. Si-on-Gr chemo-mech 지표. | 계산: V-Q 히스테리시스를 SOC 밴드별로 적분 (low band).

트렌드: 저SOC 히스테리시스: early=0.08696 → late=0.0429 (Δ =-0.04406, -0.01859V/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

고SOC 히스테리시스 (hyst_area_high)

고SOC 히스테리시스. PE activity 보조 증거. | 계산: 히스테리시스 high-SOC band 면적.

트렌드: 고SOC 히스테리시스: early=0.2146 → late=0.2348 (Δ =+0.02015, +0.0101V/100cyc) → flat · stable | aging 기대=increase | vs=stable

방전 플래토 폭 (dchg_plateau_width)

방전 플래토 구간 폭. 좁아지면 PE isolation/activity 패턴. | 계산: 방전 V 플래토 검출 후 Q 폭.

트렌드: 방전 플래토 폭: early=17.18 → late=14.39 (Δ =-2.783, -1.057Q-units/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

방전 플래토 ΔV (delta_dchg_plateau_V)

기준 대비 방전 플래토 전압 이동. | 계산: plateau_V(cycle) - plateau_V(baseline).

트렌드: 방전 플래토 ΔV : early=-0.0316 → late=-0.1041 (Δ =-0.07246, -0.03168V/100cyc) → decreasing · context | aging 기대=either | vs=context

충전 dQ/dV 피크1 V (chg_dQdV_peak1_V)

충전 IC 1번 피크 위치. FC-OCP 매칭·위상 이동 추적. | 계산: Q 보간 → SG → find_peaks.

트렌드: 충전 dQ/dV 피크1 V: early=3.483 → late=3.759 (Δ =+0.2763, +0.1079V/100cyc) → flat · context | aging 기대=either | vs=context

LAM 곡선 proxy (LAM_curve_proxy)

기준 V-Q 대비 scale 축소 proxy. 절대 LAM% 아님; 부호·bound 포화에 주의. | 계산: (1 - fit_scale) * 100 (discharge 3-param fit). scale>1 이면 음수.

트렌드: LAM 곡선 proxy: early=-2.01 → late=-12.04 (Δ =-10.03, -4.231%/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch109)

LLI 곡선 proxy (LLI_curve_proxy)

기준 대비 Q축 offset. 절대 LLI% 아님. | 계산: $\text{fit_offset} / Q_{\text{max}} \times 100$.

트렌드: LLI 곡선 proxy: early=-0.08575 → late=1.893 ($\Delta=+1.979$, +0.601%/100cyc) → increasing · context | aging 기대=either | vs=context

Rate capability factor (RCF)

0.5C 용량 / 가까운 C/3 RPT 용량. rate 손실 시 하락. | 계산: $Q_{0.5C(N)} / Q_{C/3(\text{nearest RPT})}$.

트렌드: Rate capability factor: early=0.9756 → late=0.9618 ($\Delta=-0.01379$, -0.00091791/100cyc) → flat · stable | aging 기대=decrease | vs=stable

분극 효율 비 (PER)

과전위 대비 DCIR 스케일. 수송 제한 시 변화. | 계산: $\eta(\text{SOC50}) / (\Delta I \cdot R_{\text{DCIR}_50})$.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

최대 과전위 η (eta_max)

C/3 vs 0.5C 동일 Q축 최대 전압차. | 계산: 동일 Q에서 $|V_{C3} - V_{0.5C}|$ 최댓값.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η 최댓값 SOC (eta_argmax_SOC)

η 가 최대인 SOC. 고SOC면 PE, 저SOC면 NE 쪽 힌트. | 계산: argmax_SOC of $\eta(\text{SOC})$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

PE activity 패턴 (LAM_PE_pattern_score)

NCM 이차 고립/activity 패턴. 절대 LAM% 금지. | 계산: $\text{plateau} \cdot dQdV \cdot \text{LAM_curve_proxy} \cdot dQV_{\text{log_var}}$ 가중합 (mode_weights).

트렌드: PE activity 패턴: early=0.5637 → late=0.9611 ($\Delta=+0.3974$, +0.15350-1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch109)

접촉/스택 손실 (contact_loss_score)

음 성장·분율·mech/chem 증가 증가 함. 전극 미분해. | 계산: R_{ohmic_growth} , $\Delta R\Omega$, $R\Omega_{frac}$, EoC_{10s} , mech/chem 가중합.

트렌드: 접촉/스택 손실: early=0.7834 → late=0.9914 ($\Delta=+0.208$, $+0.073020-1/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

LLI 패턴 (LLI_pattern_score)

CE·slippage·곡선 offset 기반 LLI 가설 점수. | 계산: CE ↓, CI ↑, restV/cutoff margin, LLI_curve_proxy 등.

트렌드: LLI 패턴: early=0.2637 → late=0.5959 ($\Delta=+0.3322$, $+0.12970-1/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

계면 R 패턴 (interface_R_score)

R_{ct} · τ ·VE 하락 등 계면저항 성장 가설. | 계산: R_{ct} ↑, τ_{ct} ↑, VE ↓, EoC_{60s} 등.

트렌드: 계면 R 패턴: early=0.4033 → late=0.8515 ($\Delta=+0.4483$, $+0.17430-1/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

고체확산 제한 (solid_diffusion_score)

A_{diff} ·PER·RCF 기반 확산/수송 제한. | 계산: A_{diff} ↑, PER ↑, RCF ↓ 등.

트렌드: 고체확산 제한: early=0.4528 → late=0.6434 ($\Delta=+0.1906$, $+0.036250-1/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

PE lean 점수 (PE_side_score)

$LAM_{PE_pattern}$ + feature boost + FC-OCF $\Delta hits$. | 계산: $0.75 \cdot LAM_{PE} + 0.20 \cdot PE_{boost} + peak_{boost}$.

트렌드: PE lean 점수: early=0.4775 → late=0.8151 ($\Delta=+0.3375$, $+0.13050-1/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

contact_stack lean (contact_stack_score)

contact_loss를 dominant 경쟁에 넣는 이름. $\approx contact_loss_score$. | 계산: $clip(contact_loss_score, 0, 1)$.

트렌드: contact_stack lean: early=0.7834 → late=0.9914 ($\Delta=+0.208$, $+0.073020-1/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

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NE 가설 점수 (NE_side_score)

contact × Si chemo-mech co-sign. Si 없으면 NE 확정 금지. | 계산: $0.55 \cdot \text{contact} \cdot (0.25 + 0.75 \cdot \text{si}) + 0.30 \cdot \text{NE_boost} \cdot \text{si}$.

트렌드: NE 가설 점수: early=0.1724 → late=0.2999 (Δ=+0.1276, +0.066880-1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

Si co-sign (si_cosign)

저SOC hyst·Q_relax·mech/chem·CV 등 Si chemo-mech 동시 신호. | 계산: SI_NE_COSIGN feature boost (baseline 대비 ↑ 비율).

트렌드: Si co-sign: early=0.2 → late=0.4 (Δ=+0.2, +0.12510-1/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging